Daryl BOULOU

AI Engineering Student | **Applied AI & LLM Systems**

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EXPERIENCE

EQUANS | *AI Agent Development* — **France** | *Sept. 2025 – Jan. 2026*

- Prototyped an open-source AI agent using Ollama to explore CI/CD workflow automation, adapting the approach to internal governance requirements. Delivered technical workshops on GitHub Copilot capabilities and developer workflows.
- Designed reusable prompt templates to support engineering teams with software delivery and pipeline-related tasks.
- **Tech stack**: Ollama, GitHub Copilot, GitHub Actions, Python, Git.

AQUILAB | *QA Automation Intern* — **France** | *Jan. 2025 – Apr. 2025*

- Built end-to-end automated tests with Cypress and executed functional test cycles to validate production software releases.
- Improved the application interface and implemented a product feature; contributed to full-stack development and UI/UX refinements.
- **Tech stack**: Cypress, TypeScript, JavaScript, automated testing.

EDUCATION

CESI School of Engineering | *Engineering Degree : Computer Science (AI & Data Science)* — **France** | *2024 – Present*

Politehnica University of Timişoara | *Academic Exchange Semester* — **Romania** | *Mar. 2026 – Jul. 2026*

- **Relevant coursework**: Natural Language Processing, Reinforcement Learning, Cloud Computing.

Prépa Saint-Jean | *Preparatory Classes for Engineering Schools* — **Douala, Cameroon** | *2022 – 2024*

PROJECTS

- **CameroonianLLM : Hybrid CPT & SFT Fine-Tuning of a 12B LLM** — Hugging Face | GitHub
  Fine-tuned Mistral-Nemo-Instruct (12B) with a two-stage QLoRA/rsLoRA pipeline on an NVIDIA L4: Continued Pre-Training on 23,224 texts and Supervised Fine-Tuning on 2,404 dialogues. Built a data collection and filtering pipeline processing 163K+ YouTube comments and pseudonymized conversational data; evaluated on 267 dialogues, reaching an 81.27% Camfranglais-marker rate close to human references.
- **AI Chess Coach : Local LLM & Stockfish Analysis** — GitHub
  Built a full-stack chess analysis application combining Stockfish UCI evaluations with a locally hosted Gemma model through tool calling. Grounded generated explanations in engine analysis to reduce unsupported claims; packaged a React/TypeScript frontend, FastAPI backend, Stockfish and Ollama in a multi-stage Docker build.
- **Local GitLab AI Coding Agent** — GitHub
  Engineered a Python-based autonomous coding agent integrating local LLMs with the GitLab API. Implemented issue-driven repository exploration and code modification, dedicated branch creation, commits and automated Merge Request generation.
- **Leyenda : Deep Learning Image Processing & Automated Captioning** — Live Demo | GitHub
  Built and deployed a three-stage Computer Vision/NLP pipeline: a CNN classifier (93.04% accuracy, 0.9758 ROC-AUC), a U-Net image denoiser trained with combined MSE and SSIM loss, and an InceptionV3 captioner with Bahdanau Spatial Attention and an LSTM decoder (0.750 CIDEr on MS-COCO). Deployed with Flask, Docker and Azure Container Apps.
- **MLP Language Model from Scratch** — GitHub | Article
  Reimplemented the Bengio et al. (2003) character-level MLP language model in PyTorch using raw tensor operations, Kaiming initialization and custom Batch Normalization with exponential moving statistics.

TECHNICAL SKILLS

- **Programming**: Python, TypeScript, JavaScript, SQL, Bash, HTML/CSS
- **AI/ML**: PyTorch, Hugging Face Transformers, PEFT/QLoRA, TensorFlow/Keras, Ollama, NLP, Computer Vision, Reinforcement Learning
- **Frameworks**: FastAPI, Flask, React, Cypress
- **Cloud/DevOps**: Docker, Azure Container Apps, Cloudflare, GitHub Actions, Git, GitLab CI/CD, Linux
- **Languages**: French (native), English (TOEIC: 935)